Nature Portfolio · Reporting Summary

Manuscript: Bayesian interval-censored triangulation of malignant-transformation risk in lower-grade glioma

Corresponding author: Maximilian Wartha, MD · Last updated: 27 April 2026

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting.

Statistics

For all statistical analyses, the following items are confirmed present in the figure legend, table legend, main text, or Methods section:

✓ The exact sample size ($n = 155$) and event count (77 MT, 6 non-MT deaths) are reported in the Abstract, the Cohort subsection of Results, and Methods.

✓ Measurements were taken from distinct samples (one MRI per patient at baseline; one tumour resection per patient).

✓ Statistical tests are described in Methods: Bayesian interval-censored Weibull proportional-hazards model (primary); Fine-Gray subdistribution-hazards model (cmprsk, frequentist); Cox counting-process TVC with 20-fold Rubin-pooled multiple imputation. All hypothesis tests are two-sided.

✓ All covariates are listed: contrast enhancement at diagnosis, age at diagnosis (centred at 45), tumour volume (centred), 99 PyRadiomics features (regularised-horseshoe prior in the primary fit), 3-level extent of resection, post-surgery / post-chemo / post-radiation indicators (Cox-TVC).

✓ Assumptions, corrections, and shrinkage: Bayesian shrinkage via class-wide regularised-horseshoe prior on all 102 population-level coefficients; multiple-comparison handling via the prior itself; Bonferroni correction for the two pre-specified Fine-Gray signature p-values is reported alongside the raw p.

✓ Central tendency and uncertainty: posterior median + 95 % credible interval (Bayesian) or point estimate + 95 % CI (frequentist); divergent-transition counts, R-hat, ESS_bulk and ESS_tail are reported for the Bayesian fits.

✓ For null-hypothesis tests, exact p-values are reported (e.g., Fine-Gray GLDM-FLAIR raw $p = 0.006$ / Bonferroni $p = 0.012$; era-split Fisher exact OR 2.97, $p = 0.020$; Mann-Whitney $p = 0.076$).

✓ Bayesian: priors and MCMC settings are reported in Methods (par_ratio = 0.1, scale_slab = 2, df_slab = 4, scale_global = 2; brms 2.23 / cmdstan 2.38; 4 chains × 8 000 iterations × 4 000 warmup; adapt_delta = 0.999; max_treedepth = 14).

✓ Hierarchical / complex-design accounting: Cox counting-process structure handles within-patient time-varying epochs; Rubin's rules pool across 20 multiple imputations of the interval-censored event time.

✓ Effect sizes: hazard ratios (HRs), subdistribution hazard ratios (sHRs), and odds ratios (ORs) with 95 % CIs and the data conventions used to derive them ($HR = \exp(-\alpha \cdot \beta)$ for Bayesian Weibull; AFT-to-PH conversion documented in Methods).

Software and code

Data collection

Clinical data were collected during routine care and entered into an institutional REDCap-equivalent clinical database; imaging was acquired on 19 scanner models from four manufacturers (0.4 – 3.0 T) over the 1998 – 2021 accrual window. No custom data-collection software was used.

Data analysis

All analysis code is publicly available at https://github.com/PRauch1/Malignant-transformation-in-lower-grade-glioma--a-Bayesian-interval-censored-analysis. Software versions: R 4.4 with brms 2.23, cmdstanr / cmdstan 2.38, posterior, dplyr, survival, cmprsk, mice (Rubin pooling). Python 3.10 with pandas, numpy, scipy 1.13, lifelines 0.30 (era-split sensitivity), matplotlib (figures). PyRadiomics v3.0.1 (upstream feature extraction). Deep-learning-assisted segmentation was performed with the previously published in-house pipeline (manuscript ref. 10).

Data

Data availability statement (also reported in the manuscript)

De-identified radiomic feature matrix is available from the corresponding author on reasonable request. Individual-level clinical and imaging data cannot be publicly deposited due to data-protection regulations imposed by the JKU Ethics Committee but are available from the corresponding author on reasonable request, subject to institutional and ethical approval. The 99-feature radiomic posterior-draw matrix (1000 draws × 99 features) used to render the horseshoe figures is shipped with the analysis repository.

Research involving human participants, their data, or biological material

Reporting on sex and gender

Sex was self-reported at the time of clinical encounter and recorded categorically (female/male). Sex-stratified analyses were not pre-specified for the primary endpoint; both sexes contributed to all analyses. Aggregate counts can be made available on request.

Reporting on race, ethnicity, or other socially relevant groupings

The cohort is drawn from a single academic neurosurgery centre in Linz, Austria, serving a regional population that is overwhelmingly Central European. Race or ethnicity were not collected and were not used in any analysis.

Population characteristics

155 adult-type WHO grade 2 or 3 LGG patients (77 MT; 6 non-MT deaths; 72 alive transformation-free at last follow-up). Median age at diagnosis 40.1 years (mean 41.9; range 0.7–77); 149 ≥ 18 years and 6 paediatric-age patients with adult-type histomolecular tumours. IDH-mutation status: mutant 109 (70.3 %), wildtype 42 (27.1 %), indeterminate 4 (2.6 %). Pre-operative contrast enhancement: 34 (21.9 %). Median tumour volume 35.6 cm^3 (IQR 18.4–66.2). EOR over follow-up: GTR 84 (54.2 %), STR 47 (30.3 %), Biopsy 24 (15.5 %).

Recruitment

Consecutive patients with histologically confirmed adult-type WHO grade 2 or 3 LGG accrued at our institution between 1998 and 2021 with pre-operative multiparametric MRI (T1, T1c, T2, FLAIR) and longitudinal follow-up. No external recruitment; no advertising. Selection was on the histopathological / imaging criteria prevailing at the time of accrual (WHO 2007 / 2016).

Ethics oversight

Approved by the JKU Ethics Committee (EK-2021-1042). All patients or their legal representatives provided written informed consent. The study was conducted in accordance with the Declaration of Helsinki.

Field-specific reporting

Selected: Life sciences *(clinical / observational cohort with imaging biomarkers and Bayesian survival modelling).*

Life sciences study design

Sample size

No formal a-priori sample-size calculation was performed; cohort size (n = 155) was determined by all consecutive eligible patients meeting inclusion criteria within the study period (1998–2021). Events-per-variable ratio (77 MT events / 99 candidate radiomic features ≈ 0.78) is explicitly low — the analytic strategy uses a regularised-horseshoe prior calibrated for this small-event regime (Piironen & Vehtari 2017).

Data exclusions

From 155 enrolled patients: (i) the IDH-subgroup analyses excluded 4 patients with indeterminate IDH status; (ii) the age-restricted (adult-only) sensitivity excluded 6 paediatric-age patients (n = 149); (iii) the Fine-Gray fit dropped 1 patient with a non-positive single-imputation event time (cmprsk constraint; n = 154); (iv) the WHO 2021-restricted sensitivity excluded the 19 patients with confirmed TERT promoter mutation in the absence of IDH mutation that would today be reclassified as molecular GBM (n = 136). All exclusions are pre-specified or rule-based; no patients were excluded post hoc on the basis of outcome.

Replication

All analyses are deterministic given the dataset and seeds; the analysis repository fully reproduces every reported number. Internal-validation replication: 500 patient-level bootstrap resamples (486 converged; single mid-interval imputation per replicate) for Harrell's optimism-corrected C-index. External replication in an independent multicentre cohort is the pre-registered next step.

Randomization

Not applicable. This is an observational cohort study; no randomization was performed. The era-split sensitivity (pre-2010 vs 2010+) groups patients by calendar year of diagnosis, not by random allocation.

Blinding

Tumour segmentation and PyRadiomics feature extraction were performed using the previously published deep-learning-assisted pipeline with neuroradiology review of every auto-segmentation; the segmentation/extraction step was blinded to MT outcome by virtue of being defined entirely on the baseline imaging study. All survival analyses were conducted on a pre-specified Statistical Analysis Plan locked before the present model fits were run; the two-feature radiomic signature was identified in an earlier internal analysis and pre-registered before the present Bayesian re-fit.

Reporting for specific materials, systems and methods

Materials & experimental systems

Involved in the study: Clinical data (✓). *Not involved: antibodies; eukaryotic cell lines; palaeontology / archaeology; animals; dual-use research of concern; plants.*

Methods

Involved in the study: MRI-based neuroimaging (✓). *Not involved: ChIP-seq; flow cytometry.*

Clinical data

This study is observational (not a clinical trial) and uses retrospective cohort data accordingly.

Clinical trial registration

Not applicable — this is a retrospective observational cohort study, not an interventional clinical trial. No registration number applies.

Study protocol

The Statistical Analysis Plan was locked prior to the present Bayesian model fits. The analysis pipeline (and pre-specified covariates, sensitivities, and the two-feature radiomic signature) is publicly available at the GitHub repository linked in the Code Availability statement.

Data collection

Single-institution retrospective accrual (Department of Neurosurgery, Kepler University Hospital, Johannes Kepler University Linz, Austria) of consecutive eligible patients between 1998 and 2021. Pre-operative multiparametric MRI (T1, T1c, T2, FLAIR) was acquired during routine pre-surgical workup; longitudinal surveillance MRIs were acquired at clinically indicated intervals; tissue samples (biopsy / resection) were processed by the on-site Division of Neuropathology with second-reader review of histomolecular criteria.

Outcomes

Primary endpoint: time to malignant transformation (MT), defined as histological upgrading on repeat biopsy / resection / autopsy OR radiological progression to contrast-enhancing disease with concordant clinical deterioration. Encoded as an interval [L, R] from the last pre-MT imaging to the first MT-positive imaging or transformation surgery. Competing event: death without documented MT (Fine-Gray analysis). Secondary outcomes: extent-of-resection effect on post-surgical transformation hazard; per-feature radiomic hazard ratios under regularised-horseshoe shrinkage.

Magnetic resonance imaging

Experimental design

Design type

Observational longitudinal imaging — pre-operative diagnostic MRI plus serial surveillance MRIs over follow-up.

Design specifications

Each patient contributed one pre-operative multiparametric MRI study (used for radiomic feature extraction) and a series of clinically indicated surveillance MRIs over follow-up (used to bracket the malignant-transformation event time as the interval [L, R] between the last MT-negative and first MT-positive study).

Behavioural performance measures

Not applicable — this is structural / functional anatomical imaging in a clinical oncology cohort, not a task-based functional MRI study.

Imaging type(s)

Multiparametric structural MRI: T1-weighted (T1), T1-weighted with gadolinium contrast (T1c), T2-weighted (T2), FLAIR. Perfusion MRI (rCBV) is shown for the index case in Figure 5 but was not part of the radiomics feature panel.

Field strength

Heterogeneous: 0.4 – 3.0 T. Imaging was acquired across 19 scanner models from four manufacturers over the 23-year accrual window; this heterogeneity is acknowledged in the manuscript Limitations as a source of scanner-related variance, particularly for texture features.

Sequence & imaging parameters

Acquisition was clinical-protocol-driven and varied by scanner generation and indication. Per-sequence parameters were not centrally standardised; PyRadiomics extraction used standardised pre-processing (intensity normalisation; 1-mm isotropic resampling) to mitigate inter-acquisition variance. ComBat-style harmonisation was not applied (Limitations).

Area of acquisition

Whole-brain scans; tumour-segmented volumes used for feature extraction were FLAIR (whole tumour) and T1c (enhancing sub-compartment).

Diffusion MRI

Not used.

Preprocessing

Preprocessing software

PyRadiomics v3.0.1 with intensity normalisation and 1-mm isotropic resampling. Tumour segmentation was performed using the previously established deep-learning-assisted pipeline reported in manuscript ref. 10, with neuroradiology review of every auto-segmentation.

Normalization

Intensity normalisation per PyRadiomics defaults; 1-mm isotropic resampling; per-feature z-scoring across the cohort prior to the survival models.

Normalization template

Not applicable — features are extracted in patient native space; no atlas registration was used.

Noise and artifact removal

Inter-reader segmentation reproducibility was used as the upstream feature-stability check in the originating pipeline; per-acquisition artefact removal was implicit in the DL-assisted segmentation step (segmentation review excluded any non-tumour signal).

Volume censoring

Not applicable.

Statistical modelling & inference

Model type and settings

Primary: Bayesian interval-censored Weibull proportional-hazards model with class-wide regularised-horseshoe prior across all 102 population-level coefficients (par_ratio = 0.1, scale_slab = 2, df_slab = 4, scale_global = 2). Fitted with brms 2.23 / cmdstan 2.38 (4 chains × 8 000 iterations × 4 000 warmup; adapt_delta = 0.999). Hazard ratios for covariate j computed draw-by-draw as HR = $\exp(-\alpha \cdot \beta j)$. Secondary frameworks: Fine-Gray subdistribution-hazards model (cmprsk; non-MT death as competing event); Cox counting-process TVC with 20-fold Rubin-pooled multiple imputation of the interval-censored event time.

Effect(s) tested

Pre-operative contrast enhancement; age (centred at 45); tumour volume (centred); 99 PyRadiomics features (under horseshoe shrinkage); extent of resection (GTR vs STR vs Biopsy; binary GTR vs non-GTR); post-surgery / post-chemo / post-radiation indicators (TVC).

Specify type of analysis

Whole-tumour ROI-based (FLAIR whole tumour; T1c enhancing sub-compartment). Not voxel-wise / not whole-brain.

Statistic type for inference

Posterior medians + 95 % credible intervals + Pr(HR > 1) for the Bayesian primary; point estimates + 95 % CIs + p-values for the frequentist secondaries. Optimism-corrected discrimination via Harrell's bootstrap procedure (500 patient-level resamples) for the Cox-TVC model.

Correction

Multiple-comparisons handling for the 99-feature radiomic panel is via the regularised-horseshoe prior itself (no ad-hoc adjustment); Bonferroni correction for the two pre-specified Fine-Gray signature p-values is reported alongside the raw p (raw p = 0.006 → Bonferroni p = 0.012 for GLDM-FLAIR; raw p = 0.064 → Bonferroni p = 0.128 for GLSZM-T1c). No multiple-comparisons correction was applied to the headline clinical effect (CE) because it was the single pre-specified primary covariate.

Models & analysis

Involved in the study: Multivariate modelling and predictive analysis (✓). *Not involved: functional and/or effective connectivity; graph analysis.*

Multivariate modelling and predictive analysis

Bayesian survival modelling with class-wide regularised-horseshoe prior over a 99-feature radiomic panel + 3 clinical predictors; competing-risks (Fine-Gray) and time-varying-covariate (Cox counting-process) frameworks for triangulation. Internal validation: Harrell's bootstrap-optimism procedure on the Cox-TVC fit. External validation in an independent multicentre cohort is acknowledged as the pre-registered next step.